Problem Statement:

Optimization of genetic algorithm parameter in hybrid genetic algorithm-neural network modelling: Application to spray drying of coconut milk.

In [1]:
```
import random
```

In [12]:
```
def fitness(x):
    return -(x - 5)**2 + 25
```

In [13]:
```
# Initial Population
population = [random.randint(0, 10) for _ in range(6)]
```

In [14]:
```
# Selection
def selection(pop):
    pop = sorted(pop, key=fitness, reverse=True)
    return pop[:2]   # best two
```

In [15]:
```
# Crossover
def crossover(p1, p2):
    return (p1 + p2) // 2
```

In [16]:
```
# Mutation
def mutation(child):
    if random.random() < 0.2:
        child += random.randint(-1, 1)
    return max(0, min(child, 10))  # restrict between 0-10
```

In [17]:
```
# GA Loop
generations = 5

for gen in range(generations):
    print(f"\nGeneration {gen+1}: {population}")

    p1, p2 = selection(population)
```

```
    new_population = [p1, p2]

    while len(new_population) < len(population):
        child = crossover(p1, p2)
        child = mutation(child)
        new_population.append(child)

    population = new_population
```

```
Generation 1: [7, 2, 6, 4, 6, 6]

Generation 2: [6, 4, 5, 5, 5, 5]

Generation 3: [5, 5, 5, 5, 5, 5]

Generation 4: [5, 5, 5, 5, 5, 5]

Generation 5: [5, 5, 5, 5, 5, 6]
```

In [18]:

```
# Final result
best = max(population, key=fitness)
print("\nBest solution:", best)
print("Best fitness:", fitness(best))
```

```
Best solution: 5
Best fitness: 25
```